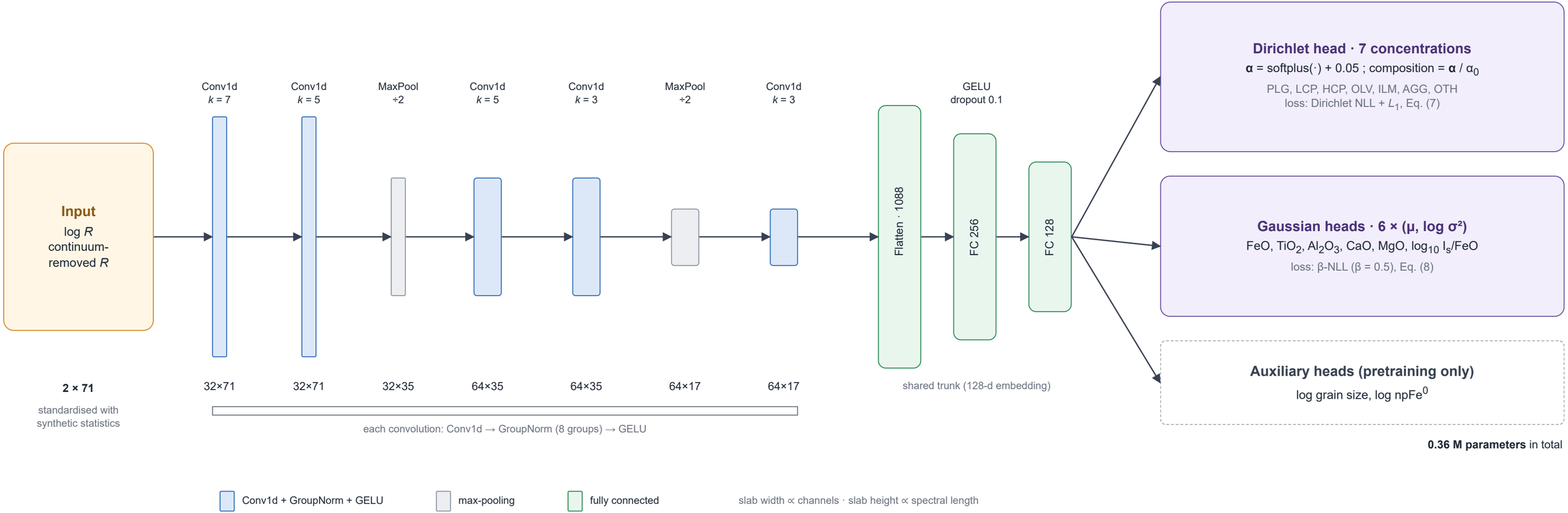

Input
log R
continuum-removed R
2 × 71
standardised with synthetic statistics
Conv1d k = 7
Conv1d k = 5
MaxPool ÷2
Conv1d k = 5
Conv1d k = 3
MaxPool ÷2
Conv1d k = 3
32×71
32×71
32×35
64×35
64×35
64×17
64×17
each convolution: Conv1d → GroupNorm (8 groups) → GELU
Flatten · 1088
GELU dropout 0.1
FC 256
FC 128
shared trunk (128-d embedding)
Dirichlet head · 7 concentrations
α = softplus(·) + 0.05 ; composition = α / α0
PLG, LCP, HCP, OLV, ILM, AGG, OTH
loss: Dirichlet NLL + L1, Eq. (7)
Gaussian heads · 6 × (μ, log σ²)
FeO, TiO2, Al2O3, CaO, MgO, log10 Is/FeO
loss: β-NLL (β = 0.5), Eq. (8)
Auxiliary heads (pretraining only)
log grain size, log npFe0
0.36 M parameters in total
Conv1d + GroupNorm + GELU
max-pooling
fully connected
slab width ∝ channels · slab height ∝ spectral length